MIL-P-36541A
16 January 1976
SUPERSEDING
MIL-P-36541
20 September 1967

MILITARY SPECIFICATION

PROCESSING MACHINE, *RADIOGRAPHIC PAPER AND DEVELOPER ASSEMBLY*

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a lightweight, field type, spring-driven, processing machine in a carrying case, for use with radiographic paper and developer assemblies in ten-second to one-minute developing. A second carrying case with a cassette, screen and grid are provided with the processing machine and are also covered herein.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-E-529	Enamel, Alkyd, Semi-Gloss.
TT-P-1757	Primer Coating, Zinc chromate, Low-Moisture Sensitivity.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-636	Boxes, Shipping, Fiberboard.

Military

MIL-M-7298	Manual, Technical Commercial Equipment.
MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-P-36148	Processing Machine, Radiographic, Paper and Developer Assembly, 110 Volt, 50-60 Cycle, AC

FSC 6525

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STANDARDS

Federal

FED-STD-595	Colors.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

25008	Processing Unit, Field (Polaroid X-Ray Film).

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 First article. When specified in the procurement document, the contractor shall furnish a sample unit for first article inspection and approval (see 4.1.2.1 and 6.2).

3.2 Material. Material shall be as specified in the Bill of Material to DPSC Drawing No. 25008.

* 3.3 Construction. The processing machine shall be a lightweight, rugged, daylight loading unit, consisting of a processor mechanism in a sheet metal container, a cassette complete with an intensifying screen and grid, and two carrying cases, one for the processor and one for the cassette and film. The entire unit shall be in strict accordance with DPSC Drawing No. 25008 and as specified herein.

* 3.3.1 Processor mechanism. The processor mechanism shall be securely fastened within the assembled sheet metal container, and the winding mechanism shall be affixed.

* 3.3.2 Sheet metal container. The container with carrying handle attached shall be finished in accordance with 3.5, then assembled with the one-minute mechanical timer, the cassette sleeve, and the packet receptacle. The container, when closed, with the cassette sleeve and the packet receptacle attached, shall be light tight as specified in 3.4.4.

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3.3.3 Cassette. The cassette design shall be in accordance with that specified in MIL-P-36148 (see 6.8).

3.3.4 Intensifying screen. Intensifying screen shall be the 10 X 12 inches industrial type (see 6.4).

3.3.5 Grid. Grid shall be the 10 X 12 inches linear type with 80 lines per inch, a focal range of 28 to 72 inches, and a 5:1 grid ratio (see 6.5).

3.3.6 Spare parts. Spare parts shall be of the quantity and type indicated in the Bill of Material. They shall be placed in an appropriate size polyethylene bag, having a self-sealing closure (see 6.6) and placed within the assembled processor.

3.3.7 Carrying cases. Both carrying cases and the foam cushions shall meet the requirements of DPSC Drawing No. 25008.

3.3.8 Interchangeability. All individual parts, or assemblies of parts of any complete unit, supplied under this specification, shall be dimensionally and functionally interchangeable with the corresponding parts or assemblies of all other like units.

3.4 Performance. Performance of the processing machine regarding roller pressure, governor accuracy, roller release, and light tightness shall be as specified herein. Developed radiographs produced by the processing machine shall not exceed the defect limits specified in table I.

3.4.1 Roller pressure. Pressure shall be such that developing chemicals spread evenly and completely from edge to edge on the radiographic surface when tested as specified in 4.4.1. Underdeveloped areas at the leading edge of the film shall not be cause for rejection provided a second radiograph developed with the same machine meets the requirements in table I.

TABLE I. Defect limits

Defect	Defect Limit
Creases and Pressure Marks	3 inches long.
Fog	Area within a 2 inch radius from a corner. Area within a 1-3/8 inch radius from any point along edge.
Streaks	Not dark enough to be mistaken as part of the image.
Scratches	1-1/4 inches from edge and not in area of the image.
Incomplete Coverage	1/2 inch on one side or 1/4 inch on both sides.

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3.4.2 Governor accuracy. Governor shall regulate the roller mechanism speed and control acceleration or faltering as indicated when tested as specified in 4.4.2.

3.4.3 Roller release. The cross-bar locking mechanism shall not release early or deposit developer on the roller surfaces under the conditions described in 4.4.3.

3.4.4 Light tightness. Light shall not leak through the material, seams, or frames of the cassette sleeve or the packet receptacle, or the joint between the sheet metal container and cover, when tested as specified in 4.4.4.

* 3.5 Finish. Finish shall be in accordance with the Bill of Material and as specified herein. All exterior surfaces, including the hinges and winding knob, the inside of the container and its cover, shall be thoroughly cleaned and chemically degreased. They then shall be given one coat of zinc-chromate primer conforming to TT-P-1757, followed by two coats of baked-on olive drab synthetic semi-gloss enamel conforming to TT-E-529. Each coat shall be sprayed on and baked in high-temperature ovens in accordance with the paint manufacturer's recommendations. The resultant finish shall be hard, smooth, and highly resistant to abrasion, chipping, and cracking. Color shall be olive drab, No. 24087 of FED-STD-595. Paint finish shall not interfere with the operation of the hinges.

3.6 Identification marking. Each processing machine shall be permanently marked with the contract or purchase order number.

3.7 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practical after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual for review and comment by the Director of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Director of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract, or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of the contract and propose to furnish manuals identical with those previously accepted then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of drafts and submission of 15 additional copies shall be at the option of the Government.

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3.8 Workmanship. Workmanship shall be first class throughout. The processing machine shall be free from defects which detract from its appearance or may impair its serviceability.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

* 4.1.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.

4.1.2.1.1 First article inspection. First article inspection shall be performed on the processor when a first article sample is required (see 3.1). This inspection shall include examinations of 4.3 and tests of 4.4.

4.1.2.1.2 Quality conformance inspection. Quality conformance inspection shall include the examinations of 4.3 and tests of 4.4.

4.1.3 Certificate of quality. Certificates of quality, supplied by the manufacturer of the component or material may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

* 4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and table II.

TABLE II. Sampling for examination

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	S-4	1.0
Total defects (major and minor combined)	S-4	4.0
For dimensional examination	S-1	2.5

4.3 Examination. The processing machine shall be examined to determine compliance with all the requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects in table III. Examination shall not be restricted to the classified possible defects listed below.

TABLE III. Classification of defects

Categories	Defects
Major	
101	Any component part missing, broken, or otherwise unusable.
102	Exposed edges not free of burs or sharpness.
103	Fabric for carrying cases not free of cuts, tears.
104	Carrying cases' stitching not 6 to 8 stitches per inch.
Minor	
201	Identification marking not complete, permanent or legible.
202	Identification marking not securely attached to unit.
203	Finish not free of chipped areas, cracks, scratches or roughness.
204	Thread not type and color specified.
205	Color of carrying cases not as specified.
206	Backstitching on the carrying cases less than 1/2 inch at each break; less than 1/4 inch where ends are not caught in other stitching.

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* 4.3.2 Dimensional examination. The processing machine shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.3 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking as required by section 5 are complied with. Defects will be as specified in table IV. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 and the AQL shall be 1.0 defects per hundred units.

TABLE IV. Examination of preparation for delivery.

Examine	Defect
Marking (exterior and interior containers)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Packaging and Packing Materials	Component missing, damaged, or otherwise defective affecting serviceability.
Workmanship	Inadequate application of component; such as incomplete closure of case liners or carton flaps, loose strapping, inadequate stapling, bulging or distortion of containers.

* 4.4 Tests. Tests shall be conducted to determine compliance with this specification. The inspection level shall be 100% for each test herein and the acceptance level shall be zero. Two or more tests shall be run concurrently wherever possible.

4.4.1 Performance. The processing machine shall be operated in accordance with manufacturer's instructions. One complete film packet shall be processed and inspected. In order to allow for possible mishandling, a second film shall be developed if the first does not pass the requirements of 3.4 and 3.4.1. During operation, the processor shall be observed for the following possible defects:

Roller speed not constant for uniform developing.
Cross-bar locking mechanism does not release at the proper instant.
Assembled functioning unit not light tight.

4.4.2 Governor accuracy. A 60 Hertz light source such as a neon light, entirely masked or otherwise shielded except for a .032 inch diameter aperture, shall be securely fastened to the processor's cross bar with the aperture location 1/4 inch from the front of the upper roller in the horizontal plane and 3/8 inch from the intersection of the upper and lower roller in the

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vertical plane when the cross bar is in the operational position. The processor shall then be operated using an unexposed film packet, properly loaded in a cassette installed in the processor. The light source shall produce 60 blips or exposures per second on a processed film pack. The exposed film, after examination for compliance with 4.4.1, 4.4.3, and 4.4.4 shall have the distance between the blips averaged in the center 4 inches of the film. The light blips of the first 3 inches and last 3 inches of the film shall then be measured and the distance between blips shall not vary more than ± 1/8 inch from the average dimension calculated from the center 4 inches of the film.

4.4.3 Roller release. Visual inspection shall be made of the film processed during the testing of 4.4.2 to insure that the cross bar locking mechanism releases at the proper instant to properly spread the chemical developer into the blotting frame at the trailing edge of the film packet.

* 4.4.4 Light tightness. The sheet metal container shall be assembled as described in 3.3.2. A lighted 100 watt incandescent lamp shall be positioned 2-1/2 inches from each of the front corners of the container. The lamps shall be at right angles to each other and equidistant horizontally and vertically from the sleeve-shrouded opening in the container. An unexposed x-ray film packet shall be processed through the machine and the film examined for evidence of light leakage.

* 5. PACKAGING

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One processing machine, as specified in 3.3, constitutes one unit of issue.

5.1.2 Level A. Each unit shall be positioned and secured within the carrying cases as specified in 3.3 and DPSC Drawing 25008.

5.1.3 Level C. Unit packaging shall be as specified in 5.1.2.

5.2 Packing. Packing shall be Level A or C as specified (see 6.2).

5.2.1 Level A:

5.2.1.1 Exterior container. Each unit shall be overboxed in an exterior container designed for a type 3 load and constructed in accordance with PPP-B-601, overseas type or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification. Fiberboard boxes shall conform to the special requirements specified in table II of PPP-B-636.

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5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

5.2.2 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The processing machine is intended for use in conjunction with NSN 6525-00-930-3275 X-Ray Apparatus, Radiographic, Medical, 2 MA, 120 PKV, 1 Tube; NSN 6525-00-930-0575 Radiographic Paper and Developer Assembly 9-3/8 by 10 inches, 25s (10 second developing); and NSN 6525-00-023-4556 Radiographic Paper and Developer Assembly 9-3/8 by 10 inches, 25s (45 second developing).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. When a first article is required (see 3.1, 4.1.2.1, and 6.3).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The contract or purchase order specifies the quantity, procedure for delivery and other specific instructions for inspection and approval of the first article.

6.4 Intensifying screen. The DuPont Cronex Industrial Intensifying Screen, 10 x 12 inch or the U.S. Radium TI-2 Industrial Intensifying Screen, 10 x 12 inch is acceptable.

6.5 Grid. The Liebel-Flarsheim Microline Stationary Grid, Catalog No. 67L0756, is acceptable.

6.6 Closure. The Kennedy "Ziplip" type closure is acceptable.

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6.7 This specification covers the following item appearing in the Federal Supply Catalog:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6525-00-930-3274	PROCESSING MACHINE, RADIOGRAPHIC PAPER AND DEVELOPER ASSEMBLY, Lightweight, field type, spring-driven

6.8 The cassette referenced in 3.3.3 appears in the Federal Supply Catalog as follows:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6525-00-601- 0600	CASSETTE, RADIOGRAPHIC PAPER AND DEVELOPER ASSEMBLY, 9-3/8 inches by 10 inches

6.9 The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DSA-DM

Project No. 6525-1120
CS

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